Journal of Monetary Economics 60 (2013) 900–916

Contents lists available at ScienceDirect

Journal of Monetary Economics

journal homepage: www.elsevier.com/locate/jme

Precautionary money demand in a business-cycle model ☆

Irina A. Telyukova [a,*], Ludo Visschers [b,c]

[a] University of California, San Diego, 9500 Gilman Drive, 0508, La Jolla, CA 92093-0508, United States
[b] Universidad Carlos III de Madrid, Spain
[c] University of Edinburgh, United Kingdom

ARTICLE INFO

Article history:
Received 22 March 2011
Received in revised form
21 August 2013
Accepted 22 August 2013
Available online 31 August 2013

Keywords:
Precautionary money demand
Velocity of money
Business cycle
Idiosyncratic risk

ABSTRACT

Precautionary demand for money is significant in the data, and may have important implications for business-cycle dynamics of velocity and other nominal aggregates. Accounting for such dynamics is a standing challenge in monetary macroeconomics: standard business-cycle models that have incorporated money have failed to generate realistic predictions in this regard. In those models, the only uncertainty affecting money demand is aggregate. We investigate a model with uninsurable *idiosyncratic* uncertainty about liquidity need. The resulting precautionary motive for holding money produces substantial improvements in accounting for business-cycle behavior of nominal variables, at no cost to real variables.

1. Introduction

In monetary macroeconomics, it is an outstanding challenge to account for business-cycle behavior of nominal aggregates and their interaction with real aggregates. Previous models that have tried to incorporate money explicitly through, for example, cash-in-advance constraints, have done so while assuming that agents face only aggregate risk, which has resulted in the demand for money being largely deterministic, so that the cash-in-advance constraint almost always binds. Such models have unrealistic implications for the dynamics of nominal variables, as well as for the interaction between real and nominal variables, when compared to the data (Cooley and Hansen, 1995; Hodrick et al., 1991).

This paper presents a theoretical and quantitative investigation of aggregate business-cycle implications of the *precautionary* demand for money. Precautionary motive for holding liquidity appears strong in the data. Telyukova (2013) documents that the median household has about 50% more liquidity than it spends on average per month, and that controlling for observables, consumption of goods requiring a liquid payment method (e.g. cash or check) exhibits volatility consistent with the presence of significant idiosyncratic risk. This risk and the resulting precautionary money demand may have important implications for aggregate money demand and other nominal variables. The goal of this paper is to investigate whether precautionary demand for money can help account for business-cycle dynamics of the velocity of money, interest rates and inflation, and their interaction with real variables.

We study the relevant mechanisms qualitatively and quantitatively in a model that combines, in each period, two types of markets in a sequential manner, and where both aggregate and idiosyncratic uncertainty are present. The first-subperiod market, termed the "credit market", is Walrasian. It is close to a standard real business-cycle model, with the production

☆ First version: June 2008.

* Corresponding author. Tel.: +1 858 822 2097.
E-mail address: itelyukova@ucsd.edu (I.A. Telyukova).

0304-3932/$ - see front matter © 2013 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.jmoneco.2013.08.006

function subject to aggregate productivity shocks, but has two distinguishing features. First, households have to decide how much money to carry out of this market for future cash consumption. Second, part of the output in the credit market is carried into the cash market by retail firms, who buy these goods on credit and subsequently transform them into cash good. This introduces an explicit link between the real and monetary sectors of the economy, as credit-market capital becomes indirectly productive in the cash market.

The second market is also competitive, but in it, agents must consume with money; this is the "cash market".[1] At the start of this subperiod, households are subject to uninsurable idiosyncratic preference shocks which determine how much of the cash goods they want to consume, but the realization of the shock is not known at the time when agents make their portfolio decisions. This generates precautionary motive for holding money. We show analytically how the idiosyncratic shocks, and the resulting heterogeneity of households, result in amplified dynamics of velocity of money, and how in their absence, the model can produce counterfactual nominal dynamics under standard parameter values.

The calibration of the model is in itself a contribution. All the existing models of the type mentioned above that have looked at aggregate behavior of nominal variables have been calibrated to aggregate data. Instead, this paper also uses household-level data on liquid consumption from the Consumer Expenditure Survey, like in Telyukova (2013), to calibrate idiosyncratic preference risk in the cash market. In general, in the few contexts where precautionary liquidity demand has appeared, it has been treated as a free parameter (e.g. Faig and Jerez, 2006). The use of household data tightly disciplines measurement of the risk, and hence of the extent of precautionary money demand.

Once calibrated, the model's equilibrium is computed to investigate the effects of real productivity shocks and monetary policy shocks. The main finding is that precautionary demand for money makes a dramatic difference for a variety of dynamic moments of nominal aggregates in the data, relative to a version of the model with the idiosyncratic risk shut down. The key factor is to break the link between aggregate money demand and aggregate consumption. In deterministic monetary business-cycle models, the cash-in-advance constraint in practice always binds, so that aggregate money demand is equivalent to aggregate cash-good consumption. This tightly links the volatilities of money demand and aggregate consumption; the latter, in turn, is not volatile enough in the data to generate observed volatility of money demand (or its inverse, velocity) and other nominal aggregates. In contrast, in the model with precautionary money demand, agents generally hold more money than they spend, so that total money demand is now linked only to the consumption of agents whose preference shock realizations make them spend all of their money in trade. Velocity of money is significantly more volatile in this heterogeneous-agent setting, thanks to the agents whose cash constraint does *not* bind, who are absent in models with only aggregate risk. In other words, idiosyncratic risk in this context does not average out in a way that can be captured by a representative agent model (see the discussion in Hodrick et al., 1991). In addition, the magnitude of idiosyncratic volatility is much higher than aggregate volatility: the standard deviation of aggregate consumption is 0.5%; the standard deviation of household-level cash consumption turns out to be around 19%.

Introducing uninsurable idiosyncratic risk into the model also changes the nature of the inflation tax, and thus has an impact on welfare costs of inflation. In any cash-credit goods model, the nominal interest rate drives a wedge between the marginal rate of transformation and the marginal rate of substitution between cash and credit goods. Without idiosyncratic shocks, this wedge affects all households equally, since all have a binding cash constraint. The cost of a positive nominal interest rate is in having to hold the money to transact in the cash market, but the allocation equates the marginal utilities across households. Instead, with uninsurable idiosyncratic risk, the additional inefficiency is that the allocation of total cash-good consumption among households is also unequal. The agents whose idiosyncratic shock realization causes their cash constraint to bind have marginal utility that is higher than that of the unconstrained agents, and thus, ex post, bear more of the cost of inflation. This raises the welfare cost of inflation. The nature of the inflation tax in the model with idiosyncratic risk also depends on whether inflation increases are anticipated. A surprise increase in inflation can exacerbate this last distortion.

These results suggest that in many monetary contexts, especially those aimed at accounting for aggregate data facts, it is important not to omit idiosyncratic uncertainty that gives rise to precautionary demand for money. As one example, we demonstrate that omitting this empirically relevant mechanism may cause the standard practice of calibrating monetary models to the aggregate money demand equation, as has been done in many cash-in-advance models and monetary search models, to produce misleading results for parameters and counterfactual quantitative implications.

This paper is related to several strands of literature. On the topic of precautionary demand for liquidity,[2] the key mechanism in this model is close to Telyukova (2013), Telyukova and Wright (2008) and Faig and Jerez (2006). In Telyukova and Wright (2008) and Telyukova (2013), the idiosyncratic uncertainty about liquidity need is shown, respectively, theoretically

[1] This setup is consistent with both cash-credit goods models in the manner of Lucas and Stokey (1987) and monetary search models in the style of Lagos and Wright (2005). In theory, money-search-style idiosyncratic matching shocks could be interpreted as a type of idiosyncratic preference shock (Wallace, 2001). However, with matching shocks, agents spend all or none of their money, while a crucial part of the argument here is that a preference shock may cause a household to spend only part of their money holdings. The natural empirical counterparts of the two types of shocks are also different.

[2] The subject of precautionary money demand goes back to at least Keynes (1936), who defined its reason as "to provide for contingencies requiring sudden expenditure and for unforeseen opportunities of advantageous purchases". Precautionary demand for money is often modeled in Baumol–Tobin-style inventory-theoretic models, from Whalen (1966) to fully dynamic stochastic models such as Alvarez and Lippi (2009). Uninsurable idiosyncratic liquidity shocks are also an essential element of models based on Diamond and Dybvig (1983). Lucas (1980) studies the equilibrium in a cash-in-advance model with precautionary demand for money.

and quantitatively, to be relevant for household portfolio decisions to hold liquid assets and credit card debt simultaneously. Faig and Jerez (2006) look at the behavior of velocity and nominal interest rates over the long run. They find that with precautionary liquidity demand, the simulated time series of velocity over the last century, interpreted as a series of steady states, fits the empirical series well. Lagos and Rocheteau (2005) study steady state properties of a monetary economy with idiosyncratic preference shocks.[3] Wang and Shi (2006) are also interested in aggregate behavior of nominal variables, but with search intensity as the key mechanism behind velocity fluctuations over the business cycle.

Section 2 below describes the model. Section 3 demonstrates *analytically* the impact of precautionary demand for money on the dynamic behavior of money, velocity and interest rates, and discusses the inflation tax. Section 4 describes the calibration strategy for several versions of the model. In Section 5 we discuss the *quantitative* role of precautionary liquidity demand, and show how omission of precautionary demand may lead model calibration and implications astray. Section 6 shows the effect of precautionary demand on welfare costs of anticipated and unanticipated inflation. Section 7 concludes.

2. Model

The economy is populated by a measure 1 of infinitely lived households, who rent labor and capital to firms, consume goods bought from the firms, and save. Two markets are open sequentially during the period. In the first subperiod, a Walrasian market opens, in which all parties involved in transactions are known and all trades can be enforced. In the second subperiod, the market is competitive, in the sense that all agents are price takers, but money is assumed essential in trade.[4] Since in the first subperiod households use cash or credit for consumption, and as discussed below, retail firms buy on credit, we will refer to this as the "credit market",[5] while the second subperiod will be termed the "cash market".

There are two types of firms in the economy. Production firms use capital and labor as inputs in production, and their output is used for consumption and capital investment in the credit market. Part of their output is also bought in the credit market by retail firms, who then transform the goods one-for-one into retail goods to be sold in the cash market.

2.1. Households

Households maximize lifetime expected discounted utility,

$$\mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t(U(c_t)-Ah_t+\vartheta_t u(q_{\vartheta,t}))\right] \tag{1}$$

where $0<\beta<1$. While here for analytical tractability the utility function is assumed separable, in computation both separable and CES utility functions will be used. Inada conditions on $U(\cdot)$ and $u(\cdot)$ ensure that consumers participate in both markets. Utility depends on consumption c_t and time spent working $h_t \in [0,1]$ in the first subperiod, and on consumption q_t and the preference shock ϑ_t in the second. First-subperiod utility follows the Hansen (1985)–Rogerson (1988) specification of indivisible labor with lotteries. The taste shock ϑ_t is realized when the credit market is already closed and money holdings can no longer be adjusted, as described below. This will lead to precautionary demand for money.

Households start the period with money holdings M_t and choose $\tilde{M}_t$ money to bring into the cash market, before ϑ_t occurs. Households also own capital k_t and nominal bonds B_t sold to them by retail firms, as detailed below. Let wage, capital rent, and the return on nominal bonds be w_t, r_t, i_{t-1}, respectively. Let the nominal price of the credit good be P_t. The budget constraint, expressed in nominal terms, is

$$M_t+(1+r_t-\delta)P_tk_t+w_tP_th_t+B_t(1+i_{t-1})=P_tc_t+\tilde{M}_t+P_tk_{t+1}+B_{t+1} \tag{2}$$

Further, given nominal price Ψ_t of the cash good, consumption $q_{\vartheta,t}$ in the cash market, conditional on the preference shock realization ϑ_t, has to satisfy $\Psi_t q_{\vartheta,t} \le \tilde{M}_t$. From now on, we will normalize household money and nominal bond holdings by the aggregate money supply $\mathcal{M}_t$, rendering them stationary, which will lead to the following normalizations: $m_t=M_t/\mathcal{M}_t$, $b_t=B_t/\mathcal{M}_t$, $p_t=P_t/\mathcal{M}_t$ and $\psi_t=\Psi_t/\mathcal{M}_t$. In addition, to formulate the budget constraint in real terms below, $\phi_t=1/p_t$ will denote the value of one unit of normalized money in terms of the credit good.

2.2. Firms

The problem of the *production* firm is standard. Given a constant returns to scale production function $y_t=e^{z_t}f(k_t,h_t)$, where z_t is the stochastic productivity level, the firm solves $\max_{k_t,h_t}\{e^{z_t}f(k_t,h_t)-w_th_t-r_tk_t\}$. The solution is characterized by the usual first-order conditions.

[3] In another broadly related paper, Hagedorn (2008) shows that strong liquidity effects can arise when precautionary demand for money is taken into account in a cash-credit good model with banking.

[4] Temzelides and Yu (2004) derive sufficient conditions under which money is essential in competitive markets. See also Levine (1991) and Rocheteau and Wright (2005).

[5] In the data, credit cards are a complex contract that for some households is a convenience device, while for others, a long-term revolving credit arrangement. Here, the credit market is a natural simplification not meant to capture credit cards tightly. Following survey data, cash consumption will be defined as goods which cannot be paid by credit.

Retail firms exist for two periods: they buy the goods in the credit market, selling nominal bonds to households to do so, sell the goods in the subsequent cash market, and settle their debt in the following credit market, before disbanding. Free entry in the retail market yields the following condition, expressed in nominal terms at time t:

$$\Pi_{rt} = \max_{q_t}\left(\frac{\psi_t q_t}{1+i_t} - \frac{q_t}{\phi_t}\right) = 0. \tag{3}$$

All cash receipts from retail sales go towards repayment next period. The repayment equals the current nominal value for the q_t goods purchased in the credit market, and is discounted using the nominal interest rate. Since the cash market is competitive, retail firms sell all their goods in equilibrium, and the cash market price is higher than the credit market price by a factor $1+i$, adjusted for marginal utility.

2.3. Monetary policy and aggregate shocks

The monetary authority follows an interest rate feedback rule:

$$\frac{1+i_{t+1}}{1+\bar{i}} = \left(\frac{1+i_t}{1+\bar{i}}\right)^{\xi_{ii}} \left(\frac{1+\pi_t}{1+\bar{\pi}}\right)^{\xi_{i\pi}} \left(\frac{y_t}{\bar{y}}\right)^{\xi_{iy}} \exp(\varepsilon_{t+1}^{mp}). \tag{4}$$

The variables with bars denote central bank's long-run target levels of output, inflation and the nominal interest rate. The term ε_t^{mp} denotes a stochastic monetary policy shock which is realized at the beginning of period t. At the end of the period, the government makes a lump sum transfer $\varpi_t \mathcal{M}_t$. The rate of money supply growth ϖ_t is adjusted by the central bank to make i_t arise as the equilibrium price.

The second, independent, aggregate shock process is on the productivity level z_t. As is standard in business-cycle literature, z_t follows an AR(1) of the form $z_{t+1} = \xi_{zz} z_t + \varepsilon_{t+1}^z$.

2.4. Recursive formulation of the household problem

From now on, we will conserve notation by omitting time subscripts, and using primes to denote $t+1$. The aggregate state variables in this economy are $S = (\mathcal{K}, z, i_{-1}, i, (1+\varpi_{-1})\phi_{-1})$: the aggregate capital stock, the technology shock, previous interest rate in the economy, current interest rate, and the previous period's post-injection real value of money, which households need to determine the current rate of inflation. The individual state variables at the beginning of the credit market are $s = (k, m, b)$: capital holdings, and money and bond holdings normalized by the money stock. The credit good is the numeraire. The household solves

$$V(k,m,b,S) = \max_{c,h,\tilde{m},k',b',\{q_\vartheta\}} \left\{ U(c) - Ah + \mathbb{E}_\vartheta\left[\vartheta u(q_\vartheta) + \beta \mathbb{E}_{z',i'} V\left(k', m', \frac{b'}{1+\varpi}, S'\right)\right]\right\} \tag{5}$$

$$\text{s.t.} \quad c + \phi\tilde{m} + k' + \phi b' = \phi m + \phi b(1+i_{-1}) + (1+r-\delta)k + wh \tag{6}$$

$$\psi q_\vartheta \le \tilde{m} \tag{7}$$

$$\pi = \frac{(1+\varpi_{-1})\phi_{-1}}{\phi} \tag{8}$$

$$m' = \frac{\tilde{m}}{1+\varpi} - \frac{\psi q_\vartheta}{1+\varpi} + \frac{\varpi}{1+\varpi} \tag{9}$$

$$z' = \xi_{zz} z + \varepsilon_1' \tag{10}$$

$$(\widehat{1+i'}) = \xi_{ii}(\widehat{1+i}) + \xi_{i\pi}\hat{\pi} + \xi_{iy}\hat{y} + \varepsilon_2' \tag{11}$$

$\hat{x}$ refers to log-deviations of the variable x from its target level. Denote the policy functions of the household's problem by $g(s,S)$, with $g_x(\cdot)$ as the policy function for the choice variable x. Assume that these functions, and the value functions, are stationary, i.e. not time-dependent.

Definition 1. A Symmetric Monetary Equilibrium is a set of pricing functions $\phi(S)$, $\psi(S)$, $w(S)$, $r(S)$; law of motion $\mathcal{K}'(S)$, value function $V(s,S)$ and policy functions $g_c(s,S)$, $g_h(s,S)$, $g_k(s,S)$, $g_b(s,S)$, $g_m(s,S)$, $\{g_{q,\vartheta}(s,S)\}$, all ϑ, such that: (i) households optimize by solving (5), given prices and laws of motion; (ii) production and retail firms optimize, as in Section 2.2; (iii) free entry of retailers implies $\Pi_r = 0$; (iv) the aggregate law of motion follows from the aggregation of all individual decisions: $\mathcal{K}'(S) = \int_0^1 g_k^i(s,S)\,di$. Finally, (v) all markets clear:

$$\int_0^1 g_m^i(s,S)\,di = 1$$

$$\int_0^1 \phi(S) g_b^i(s,S)\,di = \int_0^1 [\mathbb{E}_\vartheta g_{q,\vartheta}^i(s,S)]\,di$$

$$\int_0^1 g_h^i(s,S)\,di = H(S)$$

$$(1-\delta)\mathcal{K} + e^z f(H(S),\mathcal{K}) = \int_0^1 g_c^i(s,S)\,di + \mathcal{K}'(S) + \int_0^1 [\mathbb{E}_\vartheta g_{q,\vartheta}^i(s,S)]\,di$$

Appendix A presents the analysis of the equilibrium conditions of the problem,[6] to show that the quasi-linear specification of the utility function allows equilibria in which the distribution of wealth created by the idiosyncratic shocks in the cash market washes out in the following credit market, as long as h remains in the interior. The following result is immediate:

Result 1. *The choice of $c, \tilde{m}, k', b'$ only depends on the aggregate states S.*

Further, it can be shown that for low enough realizations of the shock ϑ, cash balances are not spent in full, and that the resulting q_ϑ is efficient, equalizing the marginal utilities of credit- and cash-good consumption. In contrast, if a shock $\overline{\vartheta}$ results in a binding cash constraint, then for any $\vartheta > \overline{\vartheta}$, the constraint will also bind. This leads to underconsumption of the cash good relative to social optimum.

What does this say about consumer payment behavior? In the data, it is reasonable to expect that different consumers make different choices with respect to how much to consume with cash versus credit. In the model, the cash market is meant to capture transactions which cannot be paid using credit. However, the "credit" market in the first subperiod is more flexible: the model is silent on whether households pay using credit or cash, as these methods can be costlessly transferred one into the other. Thus the model implicitly can accommodate the kind of heterogeneity in portfolios and payment methods seen in the data. In order to discipline the model, we will, in calibration, match the total volume of consumer transactions done by liquid payment methods (cash, check, debit) in aggregate data. Since the dynamics of interest are of aggregate variables, individual heterogeneity in payment methods will not affect them beyond getting the transaction shares right.

3. Idiosyncratic uncertainty and nominal dynamics

In this section, we demonstrate analytically that there are at least three ways in which idiosyncratic shocks to cash-good preferences can improve the quantitative performance of the model. First, the dynamic behavior of the value of money and prices varies significantly with the probability the marginal dollar is spent, i.e. that the cash constraint binds. As a result, the model with idiosyncratic shocks can accommodate values of the relative risk aversion (RRA) parameter in the standard RBC calibration range ($\sigma \in [1,4]$), whereas the model without shocks would require $\sigma < 1$ to produce realistic dynamics of prices. Second, part of velocity fluctuation is now generated in the cash market, thus increasing the overall magnitude of velocity volatility, and velocity now depends in an intuitive way on nominal interest rates. Third, the standard general-equilibrium substitution channel in cash-credit good models between cash and credit good consumption is dampened, because cash consumption will now only adjust for the binding realization of the shock. Most proofs are in Appendix B.

3.1. Dynamic behavior of real balances

The dynamic behavior of money will be an essential input for relating velocity to interest rates. It is also, however, empirically relevant in itself: one uncontroversial empirical regularity is the degree of persistence of interest rates, prices, and real balances, before and after detrending, over the business cycle. Nominal interest rates have an autocorrelation at quarterly frequency of 0.932; for real balances, it is 0.951.[7] It seems a minimal requirement that a monetary business-cycle model can replicate the sign of these autocorrelations. This requirement turns out to have important implications for the range of the RRA parameter admissible in calibration.

For the sake of exposition, assume two preference shock realizations ϑ_i, where ϑ_h leads to a binding cash constraint, and ϑ_l to a non-binding constraint. Write p for the probability of the high shock ϑ_h. Note that if $p=1$, there are no idiosyncratic shocks. We temporarily simplify the utility function in the credit market to be fully linear, $U(c)=c$. For the dynamics of real balances, the equilibrium conditions of the problem imply the relationship between real balances today ϕ and expected real balances tomorrow $\mathbb{E}\tilde{\phi}' \equiv \mathbb{E}[(\phi'(S')/(1+\varpi))U'(c')/U'(c)]$:

$$\phi = p\vartheta_h u'(\beta\mathbb{E}\tilde{\phi}')\beta\mathbb{E}\tilde{\phi}' + (1-p)\beta\mathbb{E}\tilde{\phi}'. \tag{12}$$

Recall that real balances ϕ equal normalized money supply over the price level, and since normalized money stock is 1, ϕ is also the real value of one unit of money. Eq. (12) gives that the real value of one unit of money is a weighted average of the

[6] Appendices are available on the journal website.

[7] The series are BP-filtered. Nominal interest rate is the rate on three-month treasury bonds. Real money balances are derived from M2 and the GDP deflator. Source: FRED2.

real value of a unit of money in use on the margin, needed only with probability p, and the value when this unit is not used and is carried over to the next period.

Lemma 1. *The elasticity of real balances today with respect to real balances tomorrow evaluated at an equilibrium* $\phi, \beta\mathbb{E}\tilde{\phi}'$, *is given by*

$$\varepsilon_{\phi,\beta\mathbb{E}\tilde{\phi}'} = \left(1 - \frac{1-p}{1+i}\right)(1-\sigma) + \frac{1-p}{1+i}. \tag{13}$$

Consider the case without idiosyncratic shocks, where the cash constraint always binds ($p=1$). In this case, $\varepsilon_{\phi,\beta\mathbb{E}\tilde{\phi}'} = 1-\sigma$. The value of the marginal unit of money when it needs to be spent depends on the strength of the households' motive to smooth consumption, captured by the intertemporal elasticity of substitution (IES) $1/\sigma$. Lower real value of money tomorrow implies that the value of a unit of money is also lower in today's cash market; fewer goods can be bought, raising the marginal utility. With $\sigma > 1$, households are averse enough to variation in consumption that, when real balances tomorrow are expected to decrease, the marginal utility rises so much that the demand for real balances in today's credit market goes up. This leads to a negative autocorrelation of real balances (equivalently, $\varepsilon_{\phi,\beta\mathbb{E}\tilde{\phi}'} < 0$). To avoid such counterfactual behavior of monetary variables in the setting without idiosyncratic shocks, one has to make households less averse to consumption fluctuations ($0 < \sigma < 1$). This, however, will imply an IES that is non-standard in business-cycle modeling, leading to counterfactually strong responses of real variables to real shocks. In sum, with no idiosyncratic uncertainty, one must face a tradeoff between counterfactual monetary outcomes and counterfactual real outcomes.

With idiosyncratic preference shocks, one does not have to make this unattractive choice. With $p<1$, the second term on the right-hand side of (13) is positive, and real balances tomorrow in part determine directly real balances today, because with probability $(1-p)$ the marginal unit of money is not used. Thus the elasticity of real balances today with respect to real balances tomorrow is brought closer to 1, and the demand for money is smoothed relative to the no-shock model. This means that households can have a lower IES, without implying a negative autocorrelation of real balances.[8]

3.2. Dynamic behavior of velocity

Denote by C aggregate consumption within a period. The consumption and output velocities of money in the above example with two idiosyncratic shocks can be written as

$$V_c = \frac{PC}{M} = \frac{c}{\phi} + (1-p)\frac{q_l(1+i)}{\phi} + p\frac{q_h(1+i)}{\phi}$$

$$V_y = \frac{PY}{M} = \frac{(y-(1-p)q_l - pq_h)}{\phi} + (1-p)\frac{q_l(1+i)}{\phi} + p\frac{q_h(1+i)}{\phi}$$

Looking at consumption velocity, observe that as in standard cash-in-advance and cash-credit-good models, the constrained part of the cash market always contributes 1 to the level of consumption velocity, and nothing to velocity fluctuations, because $q_h(1+i)/\phi = 1$. Thus, if $p=1$, then all velocity movement has to come from the credit market – i.e. from c or ϕ. Instead, in our model, velocity fluctuations are also created in the cash market, thanks to the low shocks where the cash constraint does not bind.[9]

One can also see this by looking at marginal rates of substitution between cash and credit market consumption. The marginal rate of substitution (MRS) for the binding and non-binding cases is

$$\frac{\vartheta_h u'(q_h)}{U'(c)} = 1 + \frac{i}{p} \tag{14}$$

$$\frac{\vartheta_h u'(q_h)}{U'(c)} = 1. \tag{15}$$

Without preference shocks ($p=1$), cash market consumption thus always depends on nominal interest rates, as in (14). Preference shocks add agents who are not constrained ($p<1$), and whose cash market consumption does not depend on i, as in (15). Having arrived in the cash market, money holdings are predetermined, and households trade off spending a dollar in the cash market versus spending it in next credit market. Given competitive pricing by retail firms', this results in an undistorted consumption allocation of unconstrained agents. Unconstrained agents do not adjust their consumption in response to changes in i, but in response to price changes, they adjust their money spending, and hence they contribute to fluctuations in velocity. Constrained agents respond to price changes through consumption, but the total amount of money spent does not move.

[8] In calibration, we will have a nominal interest rate rule with persistence. In a setting with $p=1$ and $\sigma>1$, persistence in the nominal interest rate could be achieved by alternating expansions and contractions of the money supply. Again, this would be counterfactual.

[9] Models with variable search intensity also create velocity fluctuations in the cash market (Wang and Shi, 2006). Standard search models with fixed match probabilities do not.

This analysis also sheds light on the nature of the inflation tax, relative to the standard cash-credit goods model. One can distinguish between the ex-ante inflation tax, borne in the credit market, and the ex-post tax, borne after the preference shocks are realized. The ex-ante inflation tax affects all agents equally, as long as the nominal interest rate is positive, because money is costly to hold. In a model without preference shocks, this tax is reflected in the higher cash market price than is dictated by the technology. With preference shocks, there is an additional aspect: the nominal interest rate is the cost of self-insurance against these shocks, and thus households economize on carrying real balances into the cash market. Furthermore, at higher prices in the cash market, less is sold in total, and hence less total output is produced in the credit market, which leads agents to work less in the credit market.

For the ex-post inflation tax, without preference shocks, the wedge in the MRS (14) affects everyone equally, distorting allocations relative to the optimum. With preference shocks, constrained and unconstrained agents are impacted differentially by this tax. The MRS of the unconstrained agents implies that their allocation is optimal; on their unused cash, the unconstrained agents bear the tax in the form of foregone interest. Instead, the constrained agents suffer a distortion of consumption in the cash market. Note in addition that, for a given level of q supplied by retailers, the MRS of constrained households is higher than that of the unconstrained households, and higher than the opportunity cost of holding money, $1+i$. This is contrary to the solution of a planner constrained only by q, who would equate these marginal utilities across preference shock realizations, like in the no-shock case. For the constrained agents, the cost of inflation is higher than the average in the no-shock model. We will evaluate the quantitative implications of this in the Welfare section.

Returning to velocity dynamics, the elasticity of consumption velocity with respect to i can be divided into credit-market and cash-market components $\varepsilon_{V_c,1+i} = s_c(\varepsilon_{c,1+i} - \varepsilon_{\phi,1+i}) + s_{\text{cash,nb}} \cdot \varepsilon_{\psi,1+i}$, where s_c is the share of the credit good in total consumption expenditure, and $s_{cash,nb}$ is the share of cash consumption under non-binding preference shocks.

Lemma 2. *The elasticity of the cash market price with respect to the interest rate is always positive, and is given by*

$$\varepsilon_{\psi,1+i} = \varepsilon_{(1+i)/\phi,1+i} = \frac{1+i}{\sigma(p+i)} > 0. \tag{16}$$

Thus, the less risk-averse the household is, or the smaller the probability of a binding constraint is, the more of the velocity fluctuations originate in the cash market, ceteris paribus.

To conclude the analytical discussion, we now incorporate the response of credit-good consumption c to changes in prices and interest rates. To do this, drop the linearity assumption on $U(c)$; this adds a general-equilibrium feedback effect linking nominal interest rates and velocity, through substitution between cash and credit goods. The only assumption needed for analytical tractability is that capital is constant; while this shuts down one equilibrium effect, it does not alter the other effects of interest.

Proposition 1. *The implicit elasticity of consumption velocity with respect to the nominal interest rate, caused by a one-time fully anticipated money injection (in addition to the constant rate of money growth consistent with a given steady state level of $1+i$), is*

$$\varepsilon_{V_c,1+i} = s_c\left(\frac{1}{\sigma}\frac{1+i}{p+i} - 1\right) + s_{cash,nb}\left(\frac{1+i}{p+i}\right)\left(\frac{1}{\varepsilon_{U'(c),q_h} + \sigma}\right) \tag{17}$$

The proof is in Appendix B.3.

In (17), observe the channels through which idiosyncratic uncertainty works: (i) credit market effects through the left-most term; (ii) cash market channel through the right-side $(1+i)/(p+i)$ term; and (iii) the general equilibrium substitution

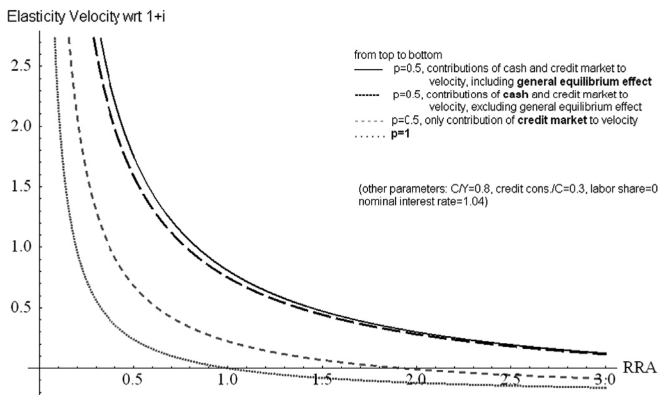

Fig. 1. Contributions of idiosyncratic uncertainty to interest rate elasticity of velocity.

channel, through $\varepsilon_{U'(c),q_h}$. We graph these components as a function of σ in Fig. 1, with $p=0.5$. Idiosyncratic shocks raise the elasticity of velocity with respect to interest rates dramatically, as signified by the vertical difference between the grey dashed and black dashed lines in the graph, and allow for a positive elasticity for a much larger range of σ. Note also that, in the difference between the top dotted and the top solid lines, the general equilibrium effect is small, but raises the elasticity further. Keeping the size of the cash market fixed, if p is lowered, the sensitivity of velocity to interest rates through the cash market channel increases.

4. Calibration

The model period is a quarter. The utility function is CES:

$$U(c,q)=\frac{((\alpha c^{\nu}+(1-\alpha)\vartheta q^{\nu})^{1/\nu})^{1-\sigma}}{1-\sigma} \tag{18}$$

We will calibrate the model both with this function and with the separable form used in most of the analytical work, which is a special case of the CES form. In that case, $U(c)=c^{1-\sigma}/(1-\sigma)$ and $u(q)=x_1 q^{1-\sigma}/(1-\sigma)$. The production function is Cobb–Douglas: $f(k,h)=k^{\theta}h^{1-\theta}$.

In the separable case, the parameters to be calibrated are β, σ, A, θ and δ, as well as x_1 and the process for the idiosyncratic shock ϑ. For the CES utility, the parameters are α, the share of credit goods in the consumption mix, and ν, the parameter that guides elasticity of substitution between cash and credit goods. Finally, the parameters of the exogenous processes $\{\xi\}$, σ_{ε_1} and σ_{ε_2} have to be calibrated.

We use the M2 measure of money supply to compute the nominal moments in the data. This choice follows the literature, e.g. Hodrick et al. (1991) and Wang and Shi (2006). Another reason is that M2 is more likely to be stationary than M1. But the most important reason for this choice is that in the data, liquid payment methods include not only cash, but also checks and debit cards, which implies inclusion in the monetary aggregate of checking accounts. In addition, it is intuitive, in studying precautionary money demand, that the monetary aggregate should include savings accounts. A concern may arise that some money in M2 earns interest, which does not happen in the model. In response to this concern, quantitative results are also presented for a modified version of the model, in which we allow interest to be paid on money that is carried over from period to period as savings by households. Appendix C presents this modification and the resulting equilibrium conditions. Notice that in this modified model, there is an extreme assumption that the representative liquid savings earn interest; in the data, a significant portion of the precautionary balances is held in non-interest-paying forms like cash and checking accounts (see Telyukova, 2013), so this approach likely overstates the share of interest-bearing liquid balances. The calibration of all the model versions is in Table 1.

Table 1
Calibrated parameters.

Parameter	Separable	CES	Separable with i^m	CES with i^m
β	0.99	0.99	0.99	0.99
σ	2	2	2	2
A	34	5	34	5
x_1	6	–	6.33	–
α	–	0.33	–	0.33
ν	–	0.39	–	0.43
θ	0.36	0.36	0.36	0.36
δ	0.02	0.02	0.02	0.02
σ_{ϑ}	0.41	0.20	0.39	0.21
ξ_{zz}	0.94	0.94	0.94	0.94
ξ_{ii}	0.78	0.78	0.78	0.78
$\xi_{i\pi}$	0.13	0.13	0.12	0.12
ξ_{iy}	0.01	0.01	0.01	0.01
ξ_{rr}	–	–	0.89	0.89
ξ_{rr}	–	–	0.01	0.01
ξ_{ry}	–	–	−0.006	−0.006
$\xi_{r\pi}$	–	–	−0.05	−0.05
σ_{ε_1}	0.006	0.006	0.006	0.006
σ_{ε_2}	0.001	0.001	0.001	0.001
σ_{ε_3}	–	–	0.0007	0.0007
$\sigma_{\varepsilon_2,\varepsilon_3}$	–	–	-7×10^{-7}	-7×10^{-7}

"Separable" – benchmark model with separable utility; "CES" – benchmark with CES utility. "With i^m" is the version of the model with interest-bearing money.

4.1. Preference and production parameters

The preference and production parameters of the model are calibrated as follows. $\beta = 0.9901$ matches the annual capital–output ratio of 3. $\sigma = 2$ is chosen within, and on the lower side of, the standard range of $[1, 4]$ in the literature. A is chosen each time to match aggregate labor supply of 0.3. The capital share of output is measured in the data to give $\theta = 0.36$. A quarterly depreciation rate of 2%, consistent with estimates in the data, gives $\delta = 0.02$.

For the separable model, the constant x_1 is calibrated to match the size of the retail (cash) market. The target is 75% of the total consumption in the model, consistent with the aggregate fact, documented in Telyukova (2013), that roughly 75% of the total value of consumer transactions in 2001 took place using liquid payments methods – cash, checks, and debit cards. This number was quoted at 82% in 1986 in Wang and Shi (2006), based on a consumer survey.

In an alternative calibration, x_1 is chosen to target the average level of M2 output velocity ($V_y = 1.897$) in the data sample (1984–2007, as detailed below). This results in the size of the cash market of 50% of total consumption. The model calibrated this way produced the same dynamic results, so they are only shown in Section 5.3 for appropriate comparison (Table 7).

For the CES utility case, the parameter x_1 is no longer used. The parameters α and ν have to be estimated jointly, together with σ_ϑ, in a simulated method of moments procedure. For α, we target the size of the cash-good market, 75%. For the parameter ν, the target is the interest rate elasticity of the ratio of aggregate cash good consumption to aggregate credit good consumption. This measure is constructed using the Consumer Expenditure Survey over the period 1980–2004.[10] Over the entire period, household non-durable consumption items are separated into cash goods and credit goods first, using the definitions discussed below for the calibration of preference shocks.[11] Then using household weights, individual cash and credit good consumption are aggregated. For the desired elasticity, the cash-to-credit log-filtered consumption ratio is then regressed on the T-Bill rate. The resulting interest rate elasticity is -1.24. The model is able to hit the targets exactly in each calibration. In the model without interest-paying money, $\nu = 0.39$ implies the elasticity of substitution between the two types of goods of 1.6.

4.2. Idiosyncratic preference shock process

Let the log of the preference shock be i.i.d. $N(0, \sigma_\vartheta)$.[12] We interpret the preference shocks as causing fluctuations in household liquid consumption *beyond* expected (e.g. seasonal or planned) fluctuations in the data. The process for this shock is calibrated following Telyukova (2013), based on quarterly data from the 2000–2002 Consumer Expenditure Survey (CEX).

The key measure is the unpredictable component of volatility of cash-good consumption in the data. This component is assumed to reflect optimal responses by households to unexpected preference shocks.[13] The standard deviation of the shock process σ_ϑ is estimated by SMM such that the standard deviation of cash-good consumption in the model matches that in the data.

The process of this measurement is described in Telyukova (2013) in detail. The first step is to separate out cash goods in the CEX data. Based on the American Bankers Association's 2004 survey of consumer payment methods, the following are defined as cash good: food, alcohol, tobacco, rents, mortgages, utilities, household repairs, childcare expenses, other household operations, property taxes, insurance, public transportation, and health insurance. Even in 2004, consumers reported paying for these types of goods with liquid assets (primarily cash and check) in 90% or more of transactions. This proportion would clearly be higher over the longer period of inquiry, 1984–2007. This definition is also conservative along some other dimensions. First, volatility of expenses could be driven by seasonality (e.g. Christmas gift shopping); to control for that in part, any expenses reported to be gifts are removed, and the regression analysis below also controls for seasonality. Second, the cash-good category excludes many situations that may be reflections of emergencies that require liquid payment, such as an emergency purchase of (or downpayment on) a durable to replace – rather than repair – a broken durable, such as a car or an appliance. Similarly, medical payments, which include co-pays or other out-of-pocket expenses, some of which are unpredictable and may require a liquid payment – are not included, because medical expenses may be payable by credit card today, even though historically this would not be the case. Thus, in measuring the volatility of cash-good consumption, using a lot of the "smooth" good categories while excluding many that may reflect emergencies other than household repairs, may understate the uncertainty that households face, against which they may hold liquid assets.

Using the above definition of cash goods, we estimate the following model:

$$\begin{aligned} \log(c_{it}^{liq}) &= \beta \mathbf{X}_{it} + u_i + \varepsilon_{it} \\ \varepsilon_{it} &= \rho \varepsilon_{i,t-1} + \eta_{it}. \end{aligned} \tag{19}$$

[10] Thanks to Dirk Krueger and Fabrizio Perri for providing us with the aggregated CEX data set for this period. All the expenditure components in their data set are deflated by the relevant component of the CPI.

[11] Clearly, in the 1980s "cash good" in the data would have been a much broader group than the 2004 definition used here. However, in the absence of survey data from that period detailing which goods were and were not cash goods, the decision of when and how to change the definition over time becomes arbitrary. The approach here is the most conservative: to keep the definition constant.

[12] Because of quasilinearity and credit markets, a shock process of the form $\vartheta' = \vartheta^\rho \varepsilon'$ with $\rho > 0$ would not change aggregate implications of the model, as can be shown from the first-order conditions of the problem.

[13] The preference shocks reflect any situation from being locked out of one's house to a significant household repair that requires payment by cash or check, e.g. In these situations, not having the money to meet the expense is very costly, which is well captured by a parameter that shifts (marginal) utility.

The vector **X** includes, depending on specification, household observables, such as age (a cubic), education, marital status, race, earnings, family size, homeownership status, as well as a set of month and year dummies. u_i is the household fixed effect. The residual ε_{it} is the idiosyncratic component of liquid consumption, consisting of persistent and transitory components. Since the preference shock is assumed i.i.d., the autoregressive component above is taken as predictable, and the innovation η_{it} as reflecting household response to the preference shocks. Table 2 presents the standard deviation of η_{it} based on the benchmark cash-good measure, and two alternatives that exclude some more predictable expense groups. The most conservative benchmark standard deviation of 19.6% is used in estimation.

The estimate of the standard deviation of the log-preference shock that results from the SMM procedure is $\sigma_\vartheta = 0.4045$ in the separable model, and 0.2042 in the CES model, where in the latter case, the parameter is estimated jointly with α and ν. The values adjust slightly if money savings are made interest-bearing in the model. The i.i.d. shock is discretized with five states using the Tauchen (1986) method, with shocks at maximum two standard deviations away from their mean. To convince the reader that we do not overstate the amount of uncertainty in expenses through the shock calibration, Fig. 2 presents the steady-state distribution of the log of liquid consumption in the model, and compares it to the empirical distribution of the log-consumption residual (η_{it}), with bins centered at the same states as in the model. It is key for the quantitative performance of the model that the probability of binding shocks is captured correctly; in this 5-state calibration, this is reflected in the top consumption state, as only the top shock binds in the calibrated equilibrium. From the figure, it is apparent that this probability is captured accurately. The discretized shock process implies that 7% of households have a binding cash constraint each period and spend all of their money. This is consistent with the findings of Telyukova (2013) that 5–7% of households have no liquid assets at a given point in time. The remaining 93% of households, in the model as in the data, are wealthy enough to hold liquid assets continually.

4.3. Aggregate shock processes

The technology and monetary policy shocks are modeled as two separate processes, as described above. For the model with no interest paid on money, these are estimated as

$$z_t = \xi_{zz} z_{t-1} + \varepsilon_1$$
$$\ln\left(\frac{1+i_t}{1+\bar{i}}\right) = \xi_{ii} \ln\left(\frac{1+i_{t-1}}{1+\bar{i}}\right) + \xi_{i\pi} \ln\left(\frac{1+\pi_{t-1}}{1+\bar{\pi}}\right) + \xi_{iy} \ln\left(\frac{y_{t-1}}{\bar{y}}\right) + \varepsilon_2$$

z_t is the Solow residual measured in the standard way, with the linear trend extracted from the Solow residual and output. The variables with bars capture long-term averages of the respective variables in the sample period, 1984–2007, as is standard in estimating central banks' targets in policy rules. The choice of years captures the period when the Federal

Table 2
Unpredictable volatility of liquid consumption, quarterly CEX data.

Measure of cash goods	Standard deviation of η_{it} (%)
Benchmark	19.6
Excluding food	27.5
Excluding food and property taxes	29.4

Standard deviation, converted into percent, of the transitory component of the residual of the regression of log-cash good consumption on household characteristics, CEX 2000–2002. See Eq. (19).

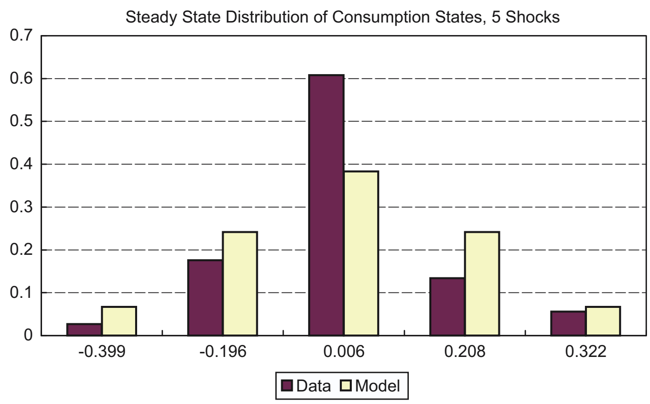

Fig. 2. Distribution of log-consumption of cash goods, data versus model discretization.

Reserve used (implicit) inflation targeting. Notice that the interest rate rule depends on endogenous variables. The interest rate variable is the Federal Funds rate.

For the model with interest-paying money, denote by R the ratio $(1+i^m)/(1+i^b)$, where i^m is the interest rate paid on money, and i^b is the interest rate on bonds. The second equation above becomes the following system, estimated by VAR using M2 own interest rate for i^m:

$$\ln\left(\frac{1+i_t^b}{1+\overline{i^b}}\right) = \xi_{ii}\ln\left(\frac{1+i_{t-1}^b}{1+\overline{i^b}}\right) + \xi_{i\pi}\ln\left(\frac{1+\pi_{t-1}}{1+\overline{\pi}}\right) + \xi_{iy}\ln\left(\frac{y_{t-1}}{\overline{y}}\right) + \varepsilon_2$$

$$\ln\left(\frac{R_t}{\overline{R}}\right) = \xi_{rr}\ln\left(\frac{R_{t-1}}{\overline{R}}\right) + \xi_{ri}\ln\left(\frac{1+i_{t-1}^b}{1+\overline{i^b}}\right) + \xi_{r\pi}\ln\left(\frac{1+\pi_{t-1}}{1+\overline{\pi}}\right) + \xi_{ry}\ln\left(\frac{y_{t-1}}{\overline{y}}\right) + \varepsilon_3.$$

5. Results

To highlight the quantitative role of precautionary demand for money, we compute, in addition to the four versions of the model above, a version of the separable benchmark with the preference shocks shut down. In this version, all agents receive the highest preference shock with probability 1; thus everyone's cash constraint always binds. This is the "no-shock model"; it closely replicates standard cash-credit goods models with only aggregate risk. The computational method is in Appendix D.

5.1. *The role of precautionary money demand*

Table 3 summarizes the dynamic properties of some key nominal variables. The first column of the table presents the data moments. The second column presents the results for the no-shock model. The last four columns show the results in the model with separable and CES utility, with and without interest-paying money. Notice that none of the result moments are targeted in calibration.

For any model version, precautionary demand for money makes a dramatic difference for the performance of the model: without it, the model is not able to capture almost any of the moments in the data, while introducing precautionary demand makes the model align successfully on nearly all of the dimensions. When mean output velocity is not a target, the model underpredicts the level of both velocities. In this model, as in other monetary business-cycle models, money turns over only once a quarter, so it is not surprising that the level is not high enough. It is also not surprising that in the no-shock model, mean velocity is higher than in the model with shocks; when the cash constraint always binds, the cash market contributes exactly 1 to velocity level every period. In the model with shocks, that contribution is less than 1 for all the non-binding shock realizations; all but 7% of the households do not spend all of the money, and hence contribute less than 1 to velocity.

Table 3
Dynamic properties of the model.

Moment	Data	No-shock model	Separable	CES	Separable with i^m	CES with i^m
$\mathbb{E}(V_y)$	1.897	1.812	1.357	1.339	1.293	1.248
$\mathbb{E}(V_c)$	1.120	1.380	1.033	1.020	0.984	0.950
$\sigma(V_y)$	0.017	0.010	0.014	0.021	0.011	0.013
$\sigma(V_c)$	0.014	0.0002	0.012	0.019	0.006	0.009
$\sigma(y)$	0.009	0.012	0.012	0.012	0.012	0.012
$\sigma(1+i^b)$	0.0026	0.001	0.002	0.0025	0.002	0.002
$corr(V_y, y)$	0.638	0.993	0.585	0.386	0.817	0.684
$corr(V_y, g_y)$	0.059	0.289	0.142	0.078	0.217	0.159
$corr(V_c, g_y)$	−0.094	0.110	−0.071	−0.065	−0.074	−0.084
$corr(V_y, g_c)$	0.127	0.539	0.233	0.109	0.242	0.464
$corr(V_c, g_c)$	−0.027	0.176	−0.155	−0.148	−0.128	−0.134
$corr(V_y, 1+i^b)$	0.714	−0.210	0.645	0. 558	0.390	0.585
$corr(V_c, 1+i^b)$	0.690	−0.896	0.897	0.648	0.912	0.915
$\varepsilon_{V_y,1+i^b}$	5.072	−1.747	4.546	4.744	2.542	4.115
$\varepsilon_{V_c,1+i^b}$	4.158	−0.123	5.072	5.046	3.363	4.653
$corr(1+\pi, 1+i^b)$	0.529	0.768	0.361	−0.032	0.572	0.465

All data logged and BP filtered. Data period: 1984–2007. Velocity moments calculated based on M2. Bond interest rate is the Fed Funds rate. Inflation measured based on GDP deflator. g_y refers to output growth. "Separable" – benchmark model with separable utility; "CES" – benchmark with CES utility. "With i^m" is the version of the model with interest-bearing money. "No-Shock" model is the version of the separable model with idiosyncratic preference shocks shut down.

If the level of velocity is targeted instead, the model matches the levels well without affecting the rest of the moments (see Table 7).

Focusing first on the models without interest-bearing money, as the analytical results suggested, the models with preference shocks do much better, in terms of volatility of velocity, than those without. This is true even with the relatively low risk aversion parameter, a parameter that needed to be high in both Hodrick et al. (1991) and Wang and Shi (2006) to begin to get significant velocity volatility. For output velocity, the separable model with the preference shocks produces 40% higher volatility than the no-shock model; for consumption velocity, the benchmark produces 60 *times* the volatility of the no-shock model. As discussed in the theory section, the reason is that with the introduction of preference shocks, the cash market contributes significantly to volatility of velocity, whereas it would contribute nothing if the cash constraint was always binding. Notice also that the model with precautionary demand gets the proportion of consumption velocity to output velocity volatility right, while it is very far off target in the no-shock model. Since consumption velocity is a major component of output velocity, consumption expenditure being 75% of GDP, the no-shock model is an unsatisfying theory of velocity dynamics because these dynamics come from the wrong source in that model.

Due to the properties of the exogenous driving process, output volatility is slightly and equally overpredicted in both the benchmark and the no-shock models. The latter, however, underpredicts volatility of velocity dramatically; thus, excess output volatility is not creating excess velocity volatility.

The model with preference shocks replicates most other moments very well too, and much better than the model without shocks. From the model analysis, we know that for $\sigma > 1$, as it is here, the correlation of velocity with nominal interest rates, and its elasticity, will be negative in the no-shock model, counter to the data. The relevant rows in the table confirm this. Instead, with the preference shocks, the relationship between velocity and nominal interest rates has the correct sign and magnitude. The correlations of output with growth of output and consumption also flip signs relative to data if the model has no idiosyncratic risk; with the shocks, the signs are correct, and the magnitudes mostly close.

The CES and separable models produce almost the same results; in the CES case, volatility of velocity is even slightly high. If the separable model is changed to have interest-bearing money, the role of precautionary demand for money is robust. The presence of interest payment on money reduces model volatility of velocity, and elasticity of velocity with respect to the bond interest rate, particularly in the separable model. This is not surprising: for unconstrained households, consumption in the cash market now responds to the movement not of the bond interest rate, but to the ratio of bond-to-money interest rates, which dampens the response of cash consumption. In addition, due to the extreme assumption that all precautionary balances earn interest, our model likely presents the lower bound on volatility of velocity. Notice, however, that with CES and interest-bearing money, we recover most of the volatility properties, as well as other dynamic properties, of the benchmark without interest-bearing money; in some dimensions, the model's performance even improves relative to data. This is a strong endorsement for the role of precautionary demand for money: not only does the model do well along the standard dimensions in the literature, where typically money is not modeled as interest-bearing and is calibrated to M2, but it passes an even higher bar of introducing interest-bearing money.

Table 4 presents the dynamic behavior of real balances (M/P), as well as autocorrelations of velocity, inflation and real balances. These moments, all close to the data in the benchmark model, show that velocity volatility does not come from excessive volatility in the value of the money stock or from volatility at the wrong frequency. Also, in the no-shock model, volatility of real balances is extremely low, again implying that it is unable to reproduce dynamics of the sources of velocity fluctuations.

5.2. Some other aggregate facts

We now assess the performance of the model against an additional set of monetary features of the business cycle, presented in Table 5. The first five columns list the performance of the model against the data, while the last two show the comparable moments from the data sample and model of Cooley and Hansen (CH, 1995). The facts are that velocity is procyclical, prices are countercyclical, and that the correlation of output and inflation with the growth of money supply is negative, with the latter being small. The model matches these facts and gets the magnitudes about right. The correlation of money growth and inflation is the only fact that is affected by introduction of interest-bearing money: the sign reverses,

Table 4
Properties of velocity volatility.

Moment	Data	No-Shock model	Separable	CES	Separable with i^m	CES with i^m
$\sigma(M/P)$	0.013	0.003	0.012	0.018	0.007	0.010
$\text{corr}(V_y, V_{y,-1})$	0.941	0.898	0.898	0.902	0.901	0.903
$\text{corr}(V_c, V_{c,-1})$	0.937	0.896	0.898	0.902	0.911	0.911
$\text{corr}(\pi, \pi_{-1})$	0.901	0.870	0.844	0.840	0.864	0.851
$\text{corr}(M/P, M/P_{-1})$	0.944	0.921	0.898	0.901	0.908	0.908

All data logged and BP filtered. Data period: 1984–2007. Velocity and money supply moments calculated based on M2. Inflation measured based on GDP deflator. "Separable" – benchmark model with separable utility; "CES" – benchmark with CES utility. "With i^m" is the version of the model with interest-bearing money. "No-Shock" model is the version of the separable model with idiosyncratic preference shocks shut down.

although in the CES case, the magnitude approaches zero. Even so, the model performs better than the CH model on all dimensions. See Appendix E for cross-correlations of some real and nominal variables with output in the model.

Finally, Table 6 highlights some aspects of the data that the model is not so successful in capturing. As the previous models in its class, the model misses the liquidity effect, i.e. the negative correlation of nominal interest rates with money growth, and prices and inflation are too flexible relative to data. The CH cash-in-advance model similarly misses these moments. The absence of the liquidity effect is not surprising, since by design, the model lacks rigidity in price adjustment.

Table 5
More monetary business cycle facts.

Moment	Data	Sep.	CES	Sep.$+i^m$	CES$+i^m$	CH data	CH
$corr(V,y)$	0.64	0.58	0.39	0.82	0.68	0.37	0.948
$corr(p,y)$	−0.13	−0.28	−0.10	−0.26	−0.28	−0.57	−0.22
$corr(g_m,y)$	−0.11	−0.15	−0.10	−0.27	−0.25	−0.12	−0.01
$corr(g_m,\pi)$	−0.06	−0.13	−0.40	0.38	0.07	−0.29	0.92

All data logged and BP filtered. Data period: 1984–2007. Velocity moments calculated based on M2. Inflation and prices measured based on GDP deflator. g_m refers to money supply growth. "Separable" – benchmark model with separable utility; "CES" – benchmark with CES utility. "With i^m" is the version of the model with interest-bearing money.

Table 6
Liquidity effect.

Moment	Data	Separable model	CH data	CH model
$corr(g_m,i)$	−0.7(M1)/0.07(M2)	0.79	−0.27	0.72
$corr(y,i)$	0.54	−0.13	0.40	−0.01
$corr(y,\pi)$	0.37	−0.25	0.34	−0.14
$corr(g_m,p)$	0.03	0.61	−0.16	0.43

All data logged and BP filtered. Data period: 1984–2007. Money supply moments calculated based on M2. Bond interest rate is the Fed Funds rate. Inflation measured based on GDP deflator. g_m refers to money supply growth. "Separable" – benchmark model with separable utility.

Table 7
No-shock model targeting money demand in the data, separable utility.

Moment	Data	Separable (target $\mathbb{E}(V_y)$)	No-shock model (target money demand)
$\mathbb{E}(V_y)$	**1.897**	**1.898**	**1.895**
$\mathbb{E}(V_c)$	1.120	1.445	1.442
$\sigma(V_y)$	0.017	0.014	0.028
$\sigma(V_c)$	0.014	0.011	0.003
$\sigma(y)$	0.009	0.012	0.022
$\sigma(c)$	0.005	0.003	0.012
$\sigma(inv)$	0.050	0.044	0.114
$\sigma(1+i)$	0.0026	0.002	0.002
$corr(V_y,y)$	0.638	0.602	0.905
$corr(V_y,g_y)$	0.059	0.145	0.411
$corr(V_c,g_y)$	−0.094	−0.070	−0.185
$corr(V_y,g_c)$	0.127	0.262	0.323
$corr(V_c,g_c)$	−0.027	−0.139	0.256
$corr(V_y,1+i)$	0.714	0.638	0.333
$corr(V_c,1+i)$	0.690	0.897	0.999
$\varepsilon_{V_y,1+i}$	**5.072**	4.469	**5.030**
$\varepsilon_{V_c,1+i}$	4.158	4.994	1.763
$corr(1+\pi,1+i)$	0.529	0.358	0.657
RRA σ		2.0	0.15

All data logged and BP filtered. Data period: 1984–2007. Velocity moments calculated based on M2. Bond interest rate is the Fed Funds rate. Inflation measured based on GDP deflator. g_y refers to output growth. Benchmark: separable model, calibrated to target $\mathbb{E}(V_y)$. "No-Shock" model is the version of the separable model with idiosyncratic preference shocks shut down. Bolded quantities represent calibration targets. RRA σ is the value of the relative risk aversion parameter needed in calibration in order to match the targets.

5.3. Pitfalls of calibration without precautionary demand

If precautionary demand for money is omitted from the model, the standard practice of calibrating monetary models to aggregate money demand (e.g. Rotemberg and Woodford (1997), Lagos and Wright (2005)), which may have a significant precautionary component in the data, may produce misleading parameter values and thus affect the quantitative performance of the model. To demonstrate this, we consider a version of the separable no-shock model where the calibration now uses two standard targets in the monetary literature: the expected value of velocity, $\mathbb{E}(V_y)$, and the elasticity of inverse velocity with respect to the nominal interest rate, $-\varepsilon_{V,1+i}$. This exercise produces different values for parameters x_1 (0.51), A (3.1) and, most importantly, the RRA parameter σ. The model without preference shocks can only reproduce the monetary targets with the value of $\sigma = 0.15$. This is not surprising: the analytical result was that the no-shock model cannot get the sign of the elasticity of velocity with respect to nominal interest rates right unless $\sigma < 1$.

The dynamic properties of this model are presented in Table 7, which compares our separable model, now calibrated to match $\mathbb{E}(V_y)$, with the no-shock model with the same target. Even when targeting $\mathbb{E}(V)$ and $\varepsilon_{V,1+i}$, the no-shock model does badly along other nominal dimensions, even for the same moments of *consumption* velocity, and now the quantitative implications on the real side of the model suffer noticeably as well. For instance, here the no-shock model doubles volatility of output, consumption and investment relative to data, while the benchmark gets these standard deviation measures fairly close to the data. In other words, even if one were willing to accept the calibrated parameters that this exercise requires, the results that the no-shock model produces are far inferior to the performance of the benchmark with preference shocks. This is true along both nominal and real dimensions, even though the no-shock model is rigged in its parameters to do well quantitatively on the nominal side.

With CES utility, one can derive the value of the parameter ν needed in order to match the target $\varepsilon_{V_c,1+i} = 4.15$, as described above. First, the elasticity of consumption velocity with respect to the nominal interest rate is given by $\varepsilon_{V_c,1+i} \approx s_{cr}(c/(c+q))\nu/(1-\nu) \approx 4.15$. With credit consumption $c/(c+q)$ at 47% of total consumption, as was the case to target the expected level of consumption velocity, this expression implies the parameter $\nu = 0.915$. If we target the credit consumption share of 25%, as in the SMM procedure described before, then we derive $\nu = 0.94$. These values of parameters imply elasticity of substitution between cash and credit goods of 11.8 and 20, respectively, for the CES model with no preference shocks to match the elasticity target.

But the model with no preference shocks also gives a closed-form implication for the parameter ν that can be estimated in the data directly. The model with CES utility implies the MRS between cash and credit goods to be $((1-\alpha)/\alpha)(q/c)^{\nu-1} = 1+i$. Taking logs and rearranging yields

$$\log\left(\frac{q}{c}\right) = -\log\left(\frac{1-\alpha}{\alpha}\right)\frac{1}{\nu-1} + \frac{1}{\nu-1}\log(1+i).$$

Based on logged and detrended data used before, the result is $\nu = 0.85$, which implies the elasticity of substitution between the two goods of 6.8. Thus the data targets once again prove impossible to match using the model with no precautionary demand. To sum up, omitting precautionary money demand from monetary models may produce inaccurate results in not only matching data facts, but also conducting policy experiments and drawing normative conclusions.

6. Welfare costs of inflation

The model is now used, in the spirit of Lucas (2000), to evaluate the welfare cost of both anticipated and unanticipated inflation.[14] As discussed above, in a model without precautionary demand for money, all agents bear the inflation tax equally: it comes from the wedge in the marginal rate of substitution between the cash and credit good created by the nominal interest rate. In the model with idiosyncratic risk, changes in steady state inflation distort cash-good consumption only for the binding shock realizations, but the constrained households value this consumption more than average, and are therefore more sensitive to these distortions.[15] Relative to a model without precautionary demand, the first channel would diminish the welfare cost of inflation, because not all agents bear it, while the second would increase it for the constrained agents. Comparing steady states, the welfare cost of inflation is the percentage reduction in consumption under the Friedman Rule that would make a household indifferent between the Friedman Rule and higher inflation. This measure is $1-\Delta$, with π_{FR} denoting inflation at the Friedman Rule:

$$U(\Delta c(\pi_{FR})) + \mathbb{E}[\vartheta u(\Delta q_\vartheta(\pi_{FR}))] - Ah(\pi_{FR}) = U(c(\pi)) + \mathbb{E}[\vartheta u(q_\vartheta(\pi))] - Ah(\pi).$$

We solve for Δ, and derive the steady state quantities in closed form (see Appendix F). The dashed line in panel (a) of Fig. 3 is the welfare cost of inflation in the no-shock model; the solid line is the cost of welfare in the model with idiosyncratic shocks.

[14] For another treatment of the welfare cost of inflation in a Bewley-type model, see also Wen (2012).

[15] In the model, the timing is such that it is the firms who take cash from one period to the next, after selling the cash goods. Given free entry of firms, the full incidence of the inflation tax is still borne by the households, because they pay a higher price for cash goods than for credit goods, adjusted for marginal utility.

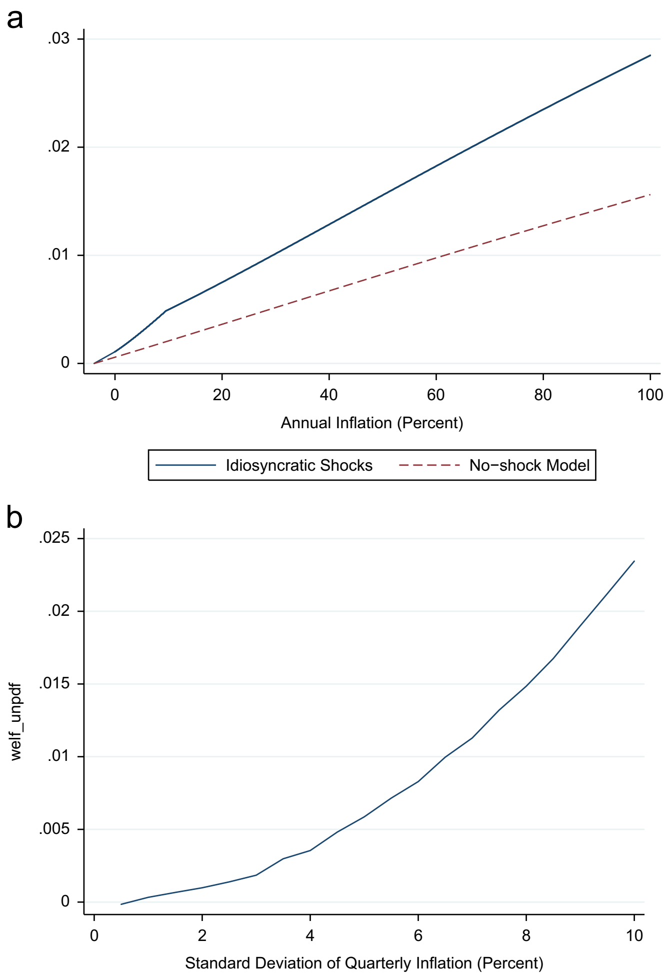

Fig. 3. Welfare cost of inflation. (a) Inflation level and (b) inflation uncertainty, 2% annual average inflation.

First, the welfare cost of 10% yearly inflation, relative to the Friedman rule, is 0.2% of the efficient level of consumption in the no-shock model, but is more than twice that, 0.5%, in the benchmark model with idiosyncratic risk. Second, the difference between the two lines is increasing with the level of annual inflation. This is because not only the cost of inflation increases for a given constrained agent as inflation rises, but in addition, the proportion of constrained agents increases with inflation. At 100% inflation, where in the benchmark model close to 30% of households are constrained, compared to only 7% at inflation rates below 9%, the welfare cost in the no-shock model is 1.6%, while in the model with shocks, it is 2.8%. Thus, a model with a full distribution of idiosyncratic shocks allows a better approximation of welfare costs at high levels of inflation than models in which the cash constraints always binds, and in which the share of consumption subject to this constraint is calibrated using velocity in low-inflation data. The underestimate of welfare costs in the no-shock model is increasing over a large range of inflation (at least up to an annual inflation of 1200%), and is relatively stable afterwards.

Regarding unanticipated inflation, in the benchmark model, there is no uncertainty about inflation within a period: the current aggregate state is revealed at the beginning of the period, households make their decisions on how much money to hold, and firms set supply of the cash good, knowing the value of money, and hence the cash-good price, in advance. One can modify the timing of the model, however, to introduce inflation “surprises”, so that households and firms make their credit-market decisions before they know the second-subperiod price level. We modify the benchmark to allow households to find out the aggregate state of the economy for period $t+1$ at the start of subperiod 2 of t. The information structure in this model is meant to be comparable to Svensson (1985); the details are in Appendix G.

This timing adjustment adds a distortion from increases in inflation, because the cash-good price ψ_t will adjust mid-period to the information regarding next period's aggregate state. Now retail firms decide on how much good q_t to take out of the credit market, and households choose money holdings, before they know ψ_t. Suppose that households observe that inflation will be higher than expected, implying lower value of money ϕ_{t+1}. The constrained households cannot increase their cash-good consumption, and can buy less in the high-shock state than they expected before. However, the supply of the cash good is fixed from the credit market, and the shortfall of demand from constrained households has to be made up by demand from unconstrained households. This means that relative to a case without unexpected inflation, the MRS of a constrained household will increase as its consumption will decline, while that of an unconstrained household will be lower than 1, so the distortion between the two types of households is exacerbated by unexpected inflation changes. Thus, an *unexpected* increase in inflation would decrease welfare by a larger amount than an *expected* increase, because q cannot adjust. As a result, a mean-preserving increase in the variance of inflation shocks can lower ex-ante welfare in this version of the model, whereas it would have no impact in the model with no inflation surprises. As panel (b) of Fig. 3 shows, welfare cost of inflation uncertainty accelerates as the standard deviation of log-inflation increases; the welfare cost of 10% standard deviation of inflation is just below 2.5% of consumption.

7. Conclusion

This study demonstrates, theoretically and quantitatively, the importance of modeling the precautionary motive for holding liquidity. By incorporating idiosyncratic expenditure risk into a standard monetary model with aggregate risk, and by carefully calibrating the idiosyncratic shocks to data, we find that the model matches many dynamic moments of nominal variables well, and greatly improves on the performance of existing monetary models without such shocks. In addition, omitting precautionary demand while targeting, in calibration, data properties of money demand – a standard calibration practice – produces inferior performance in terms of matching the data, potentially misleading implications for parameters of the model, and an understatement of welfare costs of inflation, and may therefore adversely affect the model's policy implications as well.

Acknowledgments

For helpful comments, we thank Alexander Wolman and an anonymous referee, as well as Jonathan Chiu, Makoto Nakajima, Fernando Alvarez, Manuel Amador, Paul Beaudry, Miquel Faig, Belén Jerez, Allen Head, Amartya Lahiri, Fernando Martin, Martin Schneider and Shouyong Shi as well as seminar participants at UCLA, UCSD, UC Irvine, UC Riverside, Stanford, University of Carlos III Madrid, University of Chicago, UBC, SFU, Zürich, Leicester and conference participants at EERTEC Florence, 2009 Philadelphia Fed Workshop on Monetary and Macroeconomics, LAEF Payments and Networks Conference at UCSB, CMSG Annual Meetings 2008, Chicago Fed Summer Workshop on Money, Banking and Payments, NES Moscow and IHS Vienna. We completed a previous version of this paper while visiting the Einaudi Institute for Economics and Finance. Visschers acknowledges SFU's President's Research Grant, UC3M Instituto de Economia, Project Grant ECO2010-20614 by Direccion General de Investigacion Cientifica y Tecnica (Spanish Ministry of Economic Affairs), the Juan de la Cierva Grant (Spanish Ministry of Economic Affairs), and the Bank of Spain's Programa de Investigacion de Excelencia.

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.jmoneco.2013.08.006.

References

Alvarez, F., Lippi, F., 2009. Financial innovation and the transactions demand for cash. Econometrica 77 (2), 363–402.
Cooley, T., Hansen, G., 1995. Money and the business cycle. In: Cooley, T. (Ed.), Frontiers of Business Cycle Research, Princeton University Press., pp. 175–216.
Diamond, D., Dybvig, P., 1983. Bank runs, deposit insurance, and liquidity. Journal of Political Economy 91 (3), 401–419.
Faig, M., Jerez, B., 2006. Precautionary balances and velocity of circulation of money. Journal of Money, Credit and Banking 39 (4), 843–873.
Hansen, G., 1985. Indivisible labor and the business cycle. Journal of Monetary Economics 16 (3), 309–327.
Hodrick, R., Kocherlakota, N., Lucas, D., 1991. The variability of velocity in cash-in-advance models. Journal of Political Economy 99 (2), 358–384.
Hagedorn, M., 2008. A monetary model with strong liquidity effects. Mimeo, University of Zurich.
Keynes, J.M., 1936. The General Theory of Employment, Interest and Money. Prometheus Books.
Lagos, R., Rocheteau, G., 2005. Inflation, output and welfare. International Economic Review 46 (2), 495–522.
Lagos, R., Wright, R., 2005. A unified framework for monetary theory and policy analysis. Journal of Political Economy 113 (3), 463–484.
Levine, D.K., 1991. Asset trading mechanisms and expansionary policy. Journal of Economic Theory 54 (1), 148–164.
Lucas, R., 1980. Equilibrium in a pure currency economy. Economic Inquiry 18, 203–220.
Lucas, R., 2000. Inflation and welfare. Econometrica 68, 247–274.
Lucas, R., Stokey, N., 1987. Money and interest in a cash-in-advance economy. Econometrica 55 (3), 491–513.
Rocheteau, G., Wright, R., 2005. Money in search equilibrium, in competitive equilibrium, and in competitive search equilibrium. Econometrica 73 (1), 175–202.
Rogerson, R., 1988. Indivisible labor, lotteries, and equilibrium. Journal of Monetary Economics 21 (1), 3–16.
Rotemberg, J., Woodford, M., 1997. An optimization-based econometric framework for the evaluation of monetary policy. NBER Macroeconomics Annual 12, 297–346.
Svensson, L., 1985. Money and asset prices in a cash in advance economy. Journal of Political Economy 93, 919–944.
Tauchen, G., 1986. Finite state Markov-chain approximations to univariate and vector autoregressions. Economics Letters 20, 177–181.
Telyukova, I.A., 2013. Household need for liquidity and the credit card debt puzzle. Review of Economic Studies 80 (3), 1148–1177.
Telyukova, I.A., Wright, R., 2008. A model of money and credit, with application to the credit card debt puzzle. Review of Economic Studies 75 (2), 629–647.

Temzelides, T., Yu, J., 2004. Lack-of-recall and centralized monetary trade. International Economic Review 45 (4), 1221–1227.
Wallace, N., 2001. Whither monetary economics? International Economic Review 42 (4), 847–870.
Wang, W., Shi, S., 2006. The variability of velocity of money in a search model. Journal of Monetary Economics 53, 537–571.
Wen, Yi, 2012. Liquidity and Welfare. Federal Reserve Bank of St. Louis Working Paper 2012-037A.
Whalen, E.L., 1966. A rationalization of the precautionary demand for cash. Quarterly Journal of Economics 80, 314–325.